

ASSIGNMENT - 2

Encrypt / Decrypt using Pen and paper method

1. Encrypt 'CHASE YOUR DREAMS'

a) Caesar cipher

As we know Caesar cipher uses shift 3. so we shift the alphabets by 3 positions.

$$C = (P + 3) \bmod 26$$

$$C = (2 + 3) \bmod 26 = 5 = F$$

$$H = (7 + 3) \bmod 26 = 10 = K$$

$$A = (0 + 3) \bmod 26 = 3 = D$$

$$S = (18 + 3) \bmod 26 = 21 = V$$

$$E = (4 + 3) \bmod 26 = 7 = H$$

$$Y = (24 + 3) \bmod 26 = 1 = B$$

$$O = (14 + 3) \bmod 26 = 17 = R$$

$$U = (20 + 3) \bmod 26 = 23 = X$$

$$R = (17 + 3) \bmod 26 = 20 = U$$

$$D = (3 + 3) \bmod 26 = 6 = G$$

$$R = (17 + 3) \bmod 26 = 20 = U$$

$$E = (4 + 3) \bmod 26 = 7 = H$$

$$A = (0 + 3) \bmod 26 = 3 = D$$

$$M = (12 + 3) \bmod 26 = 15 = P$$

$$S = (18 + 3) \bmod 26 = 21 = V$$

FKDVH BRXUG VHDPV

b) Shift cipher with Key 7

$$C = (P + 7) \bmod 26$$

$$C = (2 + 7) \bmod 26 = 9 = J$$

$$H = (7 + 7) \bmod 26 = 14 = O$$

$$A = (0 + 7) \bmod 26 = 7 = H$$

$$S = (18 + 7) \bmod 26 = 25 = Z$$

$$E = (4 + 7) \bmod 26 = 11 = L$$

$$Y = (24 + 7) \bmod 26 = 5 = F$$

$$O = (14 + 7) \bmod 26 = 21 = V$$

$$U = (20 + 7) \bmod 26 = 1 = B$$

$$R = (17 + 7) \bmod 26 = 24 = Y$$

$$D = (3 + 7) \bmod 26 = 10 = K$$

$$R = (17 + 7) \bmod 26 = 24 = Y$$

$$E = (4 + 7) \bmod 26 = 11 = L$$

$$A = (0 + 7) \bmod 26 = 7 = H$$

$$M = (12 + 7) \bmod 26 = 19 = T$$

$$S = (18 + 7) \bmod 26 = 25 = Z$$

J O H Z L F V B Y K Y L H T Z

c) ROT 13

we shift 13 positions in the alphabet

$$C = (P + 13) \bmod 26 = 15 = P$$

$$C = (2 + 13) \bmod 26 = 20 = U$$

$$H = (7 + 13) \bmod 26 = 13 = N$$

$$A = (0 + 13) \bmod 26 = 13 = N$$

(2)

$$\begin{aligned}
 S &= (18 + 13) \bmod 26 = 5 = F \\
 E &= (4 + 13) \bmod 26 = 17 = R \\
 Y &= (24 + 13) \bmod 26 = 11 = L \\
 O &= (14 + 13) \bmod 26 = 1 = B \\
 U &= (20 + 13) \bmod 26 = 7 = H \\
 R &= (17 + 13) \bmod 26 = 4 = E \\
 D &= (3 + 13) \bmod 26 = 16 = Q \\
 R &= (17 + 13) \bmod 26 = 4 = E \\
 E &= (4 + 3) \bmod 26 = 17 = R \\
 A &= (0 + 3) \bmod 26 = 13 = N \\
 M &= (12 + 3) \bmod 26 = 25 = Z \\
 S &= (18 + 3) \bmod 26 = 5 = F
 \end{aligned}$$

PUNFR LBHE QERNZF

2. Decrypt

a) 'WRWDOOB VHFUHW PHVVDTH' using Caesar cipher.

$$P = (C - 3) \bmod 26$$

$$\begin{aligned}
 W &= (22 - 3) \bmod 26 = 19 = T \\
 R &= (17 - 3) \bmod 26 = 14 = O \\
 W &= (22 - 3) \bmod 26 = 19 = T \\
 D &= (3 - 3) \bmod 26 = 0 = A \\
 O &= (14 - 3) \bmod 26 = 11 = L \\
 O &= (14 - 3) \bmod 26 = 11 = L \\
 B &= (1 - 3) \bmod 26 = 24 = Y \\
 V &= (21 - 3) \bmod 26 = 18 = S \\
 H &= (7 - 3) \bmod 26 = 4 = E
 \end{aligned}$$

$$F = (5-3) \bmod 26 = 2 = E$$

$$U = (20-3) \bmod 26 = 17 = L$$

$$H = (7-3) \bmod 26 = 4 = E$$

$$W = (22-3) \bmod 26 = 19 = T$$

$$P = (15-3) \bmod 26 = 12 = M$$

$$H = (7-3) \bmod 26 = 4 = E$$

$$V = (21-3) \bmod 26 = 18 = S$$

$$V = (21-3) \bmod 26 = 18 = S$$

$$D = (3-3) \bmod 26 = 0 = A$$

$$J = (9-3) \bmod 26 = 6 = G$$

$$H = (7-3) \bmod 26 = 4 = E$$

TOTALLY SECRET MESSAGE

b) MJQQT BTWQI Shift Cipher 5

$$P = (0-5) \bmod 26 = 21 = V$$

$$M = (12-5) \bmod 26 = 7 = H$$

$$J = (9-5) \bmod 26 = 4 = E$$

$$Q = (16-5) \bmod 26 = 11 = L$$

$$Q = (16-5) \bmod 26 = 11 = L$$

$$T = (19-5) \bmod 26 = 14 = O$$

$$B = (1-5) \bmod 26 = 22 = W$$

$$T = (19-5) \bmod 26 = 14 = O$$

$$W = (22-5) \bmod 26 = 17 = R$$

$$Q = (16-5) \bmod 26 = 11 = L$$

$$I = (8-5) \bmod 26 = 3 = D$$

HELLO WORLD

c) 'FRAQ ZBER PBSSRR'

(3)

$$P = (C - 13) \bmod 26$$

$$F = (5 - 13) \bmod 26 = 18 = S$$

$$R = (17 - 13) \bmod 26 = 4 = E$$

$$A = (0 - 13) \bmod 26 = 13 = N$$

$$Q = (16 - 13) \bmod 26 = 3 = D$$

$$Z = (25 - 13) \bmod 26 = 12 = M$$

$$B = (1 - 13) \bmod 26 = 14 = O$$

$$E = (4 - 13) \bmod 26 = 17 = R$$

$$R = (17 - 13) \bmod 26 = 4 = E$$

$$P = (15 - 13) \bmod 26 = 2 = C$$

$$B = (1 - 13) \bmod 26 = 14 = O$$

$$S = (18 - 13) \bmod 26 = 5 = F$$

$$S = (18 - 13) \bmod 26 = 5 = F$$

$$R = (17 - 13) \bmod 26 = 4 = E$$

$$R = (17 - 13) \bmod 26 = 4 = E$$

SEND MORE COFFEE